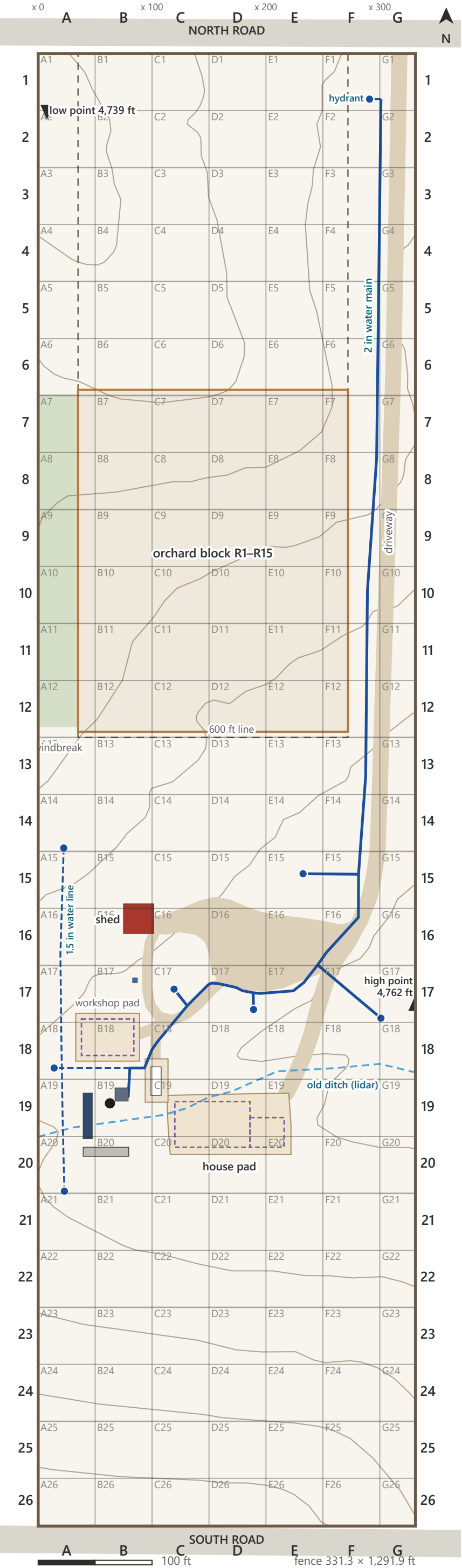


Mountain Prairie Acres: Site Master Grid

One coordinate system for every project on the property: lot feet from the northwest fence corner, 50 ft cells A–G by 1–26, tied to latitude and longitude. Pages and code helpers all read the same data file.

Whole lot, north up, 50 ft cells



Reading a grid reference

The diagram shows a 50x50 ft cell with a point at (63, 921). The origin (0,0) is the northwest fence corner. The cell is labeled x63, y921.

Lot feet: x is feet east of the west fence, y is feet south of the north fence. The origin (0, 0) is the northwest fence corner, and the fences run true north–south and east–west.

Cells: a column letter (A–G, west to east) plus a row number (1–26, north to south). Each cell is 50 × 50 ft, so *B19* is x 50–100, y 900–950. Write a spot as *B19* (x 63, y 921).

Lot and location

FENCED LOT

331.3 × 1,291.9 ft

9.83 acres inside the fence

LIKELY PARCEL

330 × 1,320 ft (10 ac)

to the road centerlines; not surveyed

ORIGIN (0, 0)

43.47545, -115.59484

northwest fence corner (NAD83 / GPS)

DRIVEWAY ENTRANCE

43.47550, -115.59364

open in Google Maps (for directions)

LOT AXES

x runs 89.9° true

within 0.2° of due east; y is due south

POSITION ACCURACY

about 10 ft

USGS NAIP-based; a shed GPS pin checks within 5 ft

Feature coordinates

Feature (id)	Cells	Center x, y	Size, ft	Ground, ft	Lat, long
Boundary and access					
Fence (lot line) fence	A1–G26	166, 646	331.3 × 1,291.9	—	43.47368, -115.59421
North road road_north	off grid	170, -18	24 wide	—	43.47550, -115.59420
South road road_south	off grid	170, 1,305	24 wide	—	43.47187, -115.59419
Driveway entrance driveway_entrance	off grid	318, -18	—	—	43.47550, -115.59364
9 · Driveway driveway	G1 → E19	306, 455	14 wide	4,752.8	43.47420, -115.59368
9 · Yard and turnaround yard	C15–F17	193, 780	8,400 sq ft	4,759.0	43.47331, -115.59411
9 · Loop drive around the island loop_path	C16 → D17	102, 839	9 wide	4,758.4	43.47315, -115.59445
9 · Path to the workshop pad path_workshop	C17 → B18	101, 855	10 wide	4,758.8	43.47311, -115.59445
9 · Path to the RV pad path_rv	C18	103, 866	9 wide	4,759.2	43.47308, -115.59444
Homestead					
1 · Red shed shed	B15–C16	88, 759	26.5 × 26	4,756.2	43.47337, -115.59450
2 · Outhouse outhouse	B17	85, 813	4 × 4	4,757.7	43.47322, -115.59451
3 · Solar container container_solar	A19, A20	44, 932	8 × 40	4,757.9	43.47290, -115.59467
3 · Storage container container_storage	A20, B20	60, 964	40 × 8	4,758.3	43.47281, -115.59461
4 · House pad house_pad	C19–E20	168, 939	109 × 54	4,759.8	43.47288, -115.59420
5 · Irrigation tank irrigation_tank	B19	63, 921	9 across	4,758.7	43.47293, -115.59459
6 · Pump house pump_house	B19	73, 913	11.5 × 11.5	4,759.1	43.47295, -115.59456
7 · Workshop pad workshop_pad	A17–B18	61, 863	56 × 42	4,758.4	43.47308, -115.59460
8 · RV pad rv_pad	B18–C19	104, 901	20 × 38	4,759.5	43.47298, -115.59444
8 · RV camper rv	B18–C19	104, 902	9 × 25	4,759.5	43.47298, -115.59444
Planned buildings					
4 · House (planned) house	C19–D20	153, 940	66 × 40	4,759.6	43.47288, -115.59425
4 · Garage (planned) garage	D19–E20	201, 947	30 × 26	4,760.2	43.47286, -115.59407
7 · Workshop (planned) workshop	A17–B18	61, 863	46 × 32	4,758.4	43.47308, -115.59460
Utilities					
Water main, 2 in PVC water_main	B19 → G1	288, 546	2 in pipe, 1,015 long	4,755.4	43.47396, -115.59375
Water line west to the fence, 1.5 in PVC water_tee_west	A18, B18	47, 890	1.5 in pipe, 65 long	4,758.6	43.47301, -115.59465
Water line along the west fence, 1.5 in water_line_west	A14 → A20	20, 847	1.5 in pipe, 300 long	4,757.3	43.47313, -115.59476
Water branch to the island hydrant, 2 in water_branch_island	C17	125, 828	2 in pipe, 20 long	4,758.7	43.47318, -115.59436
Water branch to the hydrant south of the yard, 2 in water_branch_yard_south	D17	189, 831	2 in pipe, 15 long	4,759.9	43.47317, -115.59412
Water branch to the east hydrant, 2 in water_branch_east	E16 → G17	273, 823	2 in pipe, 75 long	4,760.9	43.47320, -115.59380
Water branch to the hydrant north of the yard, 2 in water_branch_yard_north	E15, F15	257, 720	2 in pipe, 50 long	4,758.9	43.47348, -115.59386
Water lateral at the north end, 1.5 in water_lateral_north	F1, G1	296, 40	1.5 in pipe, 10 long	4,748.5	43.47534, -115.59372
Hydrant, west end of the 1.5 in line hydrant_west	A18	14, 890	—	4,757.9	43.47301, -115.59478
Hydrant, south end of the west fence line hydrant_fence_south	A20	23, 998	—	4,757.6	43.47271, -115.59474
Hydrant, north end of the west fence line hydrant_fence_north	A14	22, 697	—	4,752.6	43.47354, -115.59475
Hydrant, in the island hydrant_island	C17	119, 821	—	4,758.6	43.47320, -115.59438
Hydrant, south of the yard hydrant_yard_south	D17	189, 839	—	4,760.0	43.47315, -115.59412
Hydrant, near the east fence hydrant_east	G17	301, 846	—	4,761.5	43.47313, -115.59370
Hydrant, north of the yard hydrant_yard_north	E15	233, 720	—	4,758.5	43.47348, -115.59396
Hydrant, north end hydrant_north	F1	291, 40	—	4,748.3	43.47534, -115.59374
Orchard					
Orchard strip orchard_area	A1–F12	154, 300	237 × 600	4,744.1	43.47463, -115.59426
Orchard block R1-R15 orchard_block	A6–F12	154, 445	237 × 300	4,750.3	43.47423, -115.59426
Windbreak windbreak	A6–A12	18, 445	35 × 292	4,748.0	43.47423, -115.59477
Terrain					
High point high_point	G17	331, 835	—	4,761.7	43.47316, -115.59358
Low point low_point	A2	0, 50	—	4,739.2	43.47532, -115.59484
Old ditch line old_ditch	A20 → G18	165, 912	8 wide	4,759.6	43.47295, -115.59421

Using the grid in code

Web pages load data/site_grid.js (grid, features, latitude/longitude tie) and data/site_terrain.js (lidar elevations), then call SiteGrid:

```
<script src="data/site_grid.js"></script>
<script src="data/site_terrain.js"></script>

SiteGrid.cellOf(63, 921)           // 'B19'
SiteGrid.cellBounds('B19')         // { x0: 50, y0: 900, x1: 100, y1: 950 }
SiteGrid.center('pump_house')      // [73.25, 913.25]
SiteGrid.cellsOf('house_pad')      // ['C19', 'D19', 'E19', 'C20', 'D20', 'E20']
SiteGrid.elevationFt(63, 921)      // 4758.7 (feet above sea level)
SiteGrid.toLatLon(63, 921)         // [43.472925, -115.594594]
SiteGrid.fromLatLon(43.473383, -115.594507) // [86.5, 754.3] (a GPS pin on the shed)
SiteGrid.treeXY(5, 3)              // [74, 505] (orchard tree R5-C3, cell B11)
SiteGrid.mapsLink(63, 921)         // Google Maps link to that spot
```

Scripts and the command line use tools/site_grid.py, which reads the same files:

```
python tools/site_grid.py cell 63 921           # B19
python tools/site_grid.py where B19             # cell bounds and the features in it
python tools/site_grid.py feature pump_house     # center, size, cells, ground, lat/long
python tools/site_grid.py dist pump_house house  # gap and ground change
python tools/site_grid.py tree 5 3              # orchard tree R5-C3: x, y, cell, ground
python tools/site_grid.py fromlatlon 43.473383 -115.594507
from site_grid import SiteGrid; grid = SiteGrid(); grid.cell_of(63, 921); grid.elevation_ft(63, 921)
```

Conventions

- Feet throughout. Elevations are above sea level (NAVD88) from USGS 3DEP lidar flown before the homestead pads were graded.
- Status is existing or planned; planned buildings move once their spots are staked. To add or move something, edit the JSON between the BEGIN and END markers in docs/data/site_grid.js (rect x0/y0/x1/y1, circle cx/cy/r, polygon or line pts, point x/y), then run python -m unittest discover -s tools.
- Latitude and longitude come from the fitted lot frame, good to about 10 ft: fine for maps and directions, not for boundary questions. For precise layout, measure from the fences.
- Buried water lines (2 in PVC solid, 1.5 in dashed) and the 8 frost-free hydrants (blue dots) are traced from the owner's hand-drawn map (October 2026): good to about 5 ft around the homestead and 10 ft at the north end, so dig by hand near them.